

Data Warehouse

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Introduction.

In today's data-driven world, global supply chain and logistics companies generate enormous volumes of transactional data across ordering, inventory, customer management, shipping, and delivery operations. Harnessing this data for strategic decision-making requires robust data integration, transformation, and analytical capabilities. Data Warehousing and Business Intelligence (DWBI) solutions provide the foundation for converting raw, distributed, and heterogeneous data into structured, meaningful insights that support operational efficiency, cost reduction, and customer satisfaction.

This assignment focuses on designing and implementing a complete data warehousing solution for the **DataCo Smart Supply Chain** system. The goal is to simulate a real-world global supply chain scenario where data is sourced from both structured and semi-structured systems—namely, a normalized OLTP database (containing orders, products, shipments, etc.) and a flat CSV file (customer source data)—mimicking the common challenges faced in modern enterprise environments.

The assignment is structured into multiple stages: dataset selection and understanding, source data preparation, solution architecture design, dimensional modeling using a star schema, and ETL development using SQL Server Integration Services (SSIS). The data warehouse is designed to support future OLAP analysis and reporting capabilities, with key considerations such as Slowly Changing Dimensions (Type 2 for Product and Customer), derived column transformations, and accumulating fact table updates built into the ETL logic.

By completing this project, key data warehousing concepts were practically applied, including data staging, data cleansing, lookup transformations, surrogate key generation, Slowly Changing Dimension handling, and dimension-fact integration. The final deliverable is a fully functional data warehouse schema that supports analytical queries for monitoring sales performance, shipping efficiency, late delivery risks, and overall supply chain operations over time.

This hands-on experience demonstrates the critical role of data warehousing in transforming raw supply chain data into actionable business intelligence, which can ultimately support strategic planning, operational optimization, and competitive advantage in the logistics industry.

Step 1: Data Set Selection

1.1 Introduction

The dataset selected for this assignment is the **DataCo Smart Supply Chain for Big Data Analysis** (available on Kaggle). It represents a realistic, large-scale supply chain and logistics operation of a global e-commerce company named **DataCo Global**. The data simulates realworld transactional activities across provisioning, production, sales, and commercial distribution.

1.2 Dataset Overview

Key Characteristics of the Dataset:

- **Size:** Approximately 180,000+ records (transactions).
- **Time Period:** Covers multiple years of supply chain operations (primarily 2015–2018), satisfying the requirement of at least one year of meaningful data.
- **Domain:** Global supply chain management, including order processing, product catalog, customer information, shipping, and delivery performance.
- **Main File:** Originally provided as a single CSV file (DataCoSupplyChainDataset.csv) with 52–55 columns.
- **Product Types:** Clothing, Sports, and Electronic Supplies.

1.3 Source Types and Formats

sources:

Source Type	Example Files/Tables	Description
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SQL Database	DataCo_DataBase	Contains normalized tables: Department, Category, Product, Order, Order_Item, and Shipment.
Flat File	Customer_Source.csv	containing customer demographic and location Details.

This separation creates heterogeneous data sources (SQL + CSV), demonstrating real enterprise integration challenges.

1.4 Coverage and Realism

The dataset was designed to include sufficient variety and depth to support key data warehousing requirements:

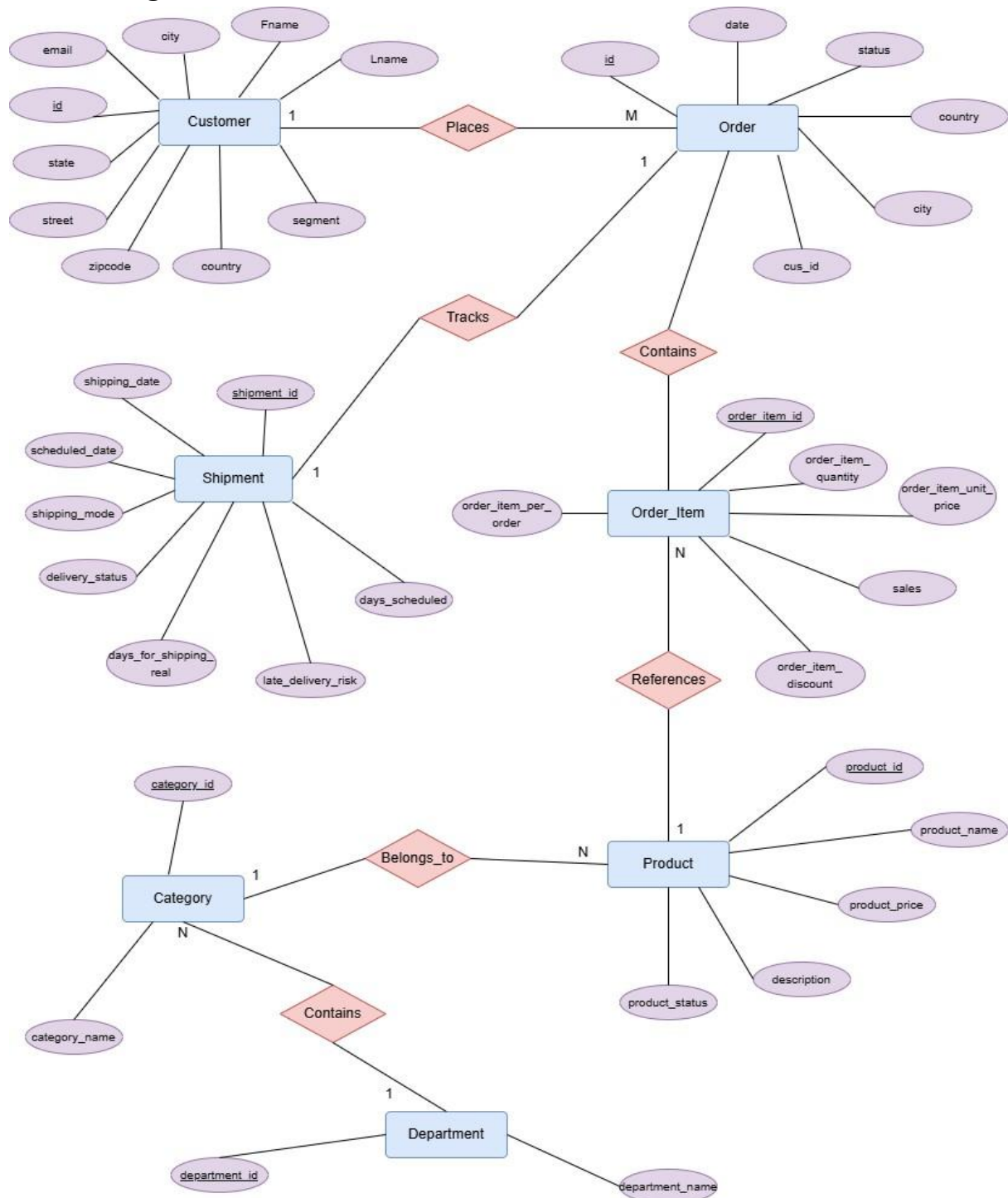
- **Time coverage:** Multiple years of real supply chain operations (primarily 2015–2018) with full-year transactional data.
- **Diverse product categories:** Clothing, Sports, and Electronic Supplies.
- **Order scenarios:** Various order statuses, customer segments (Consumer, Corporate, Home Office), and discount levels.
- **Shipping and delivery performance:** Four shipping modes (First Class, Same Day, Second Class, Standard Class), real vs scheduled shipping days, and late delivery risk indicators.
- **Multiple transaction types:** High-volume order items with unit prices, discounts, and sales amounts.
- **Geographically diverse customer data:** Customers from multiple countries, states, and cities worldwide.

1.5 Suitability for Data Warehousing and BI

The dataset satisfies all major requirements specified in the assignment:

Requirement	Details
OLTP structure	Normalized database with transactional records
Novel scenario	Global supply chain and logistics dataset not used in lab work
One year of data	Covers multiple years of supply chain transactions (2015–2018)
Multiple source types	SQL Server database + external CSV file
Complex ETL support	Includes SCD Type 2, derived columns, and accumulating facts
Warehouse-ready dimensional modeling	Supports star schema with fact and multiple dimension tables
SSAS cube and reporting compatibility	Hierarchies available (e.g., Department → Category → Product)

1.6 ER Diagram



Step 2: Source Data Description

To simulate a realistic enterprise data warehousing environment, the source data for this project was extracted from two distinct systems: a structured SQL Server database and an unstructured flat file. These heterogeneous sources were selected to demonstrate the ETL process's ability to handle varying data formats while ensuring integration accuracy and consistency.

2.1 SQL Server Database Source

The primary source system is a normalized OLTP-style SQL Server database named **DataCo_SupplyChain_SourceDB**. This database represents the core transactional operations of DataCo Global and contains the following interrelated tables:

- **Department** – Contains department information (e.g., Footwear, Apparel, Electronics).
- **Category** – Represents product categories linked to departments.
- **Product** – Stores product master data including name, price, and status.
- **Order** – Holds order header information such as order date, status, and customer link.
- **Order_Item** – Captures detailed order line items with quantity, unit price, discount, and sales amount.
- **Shipment** – Contains shipping and delivery details including shipping mode, delivery status, and late delivery risk.

All tables maintain referential integrity through well-defined primary and foreign key relationships. Data was extracted using SQL queries and loaded into staging tables via SSIS.

2.2 Flat File Source

In addition to the structured database, an external flat file named **Customer_Source.csv** was used to simulate customer data originating from a separate customer relationship management (CRM) system or third-party source.

This CSV file contains customer demographic and location information with the following key fields:

- Customer ID
- Customer First Name and Last Name
- Customer Segment (Consumer, Corporate, Home Office)
- Customer City, State, Country, and Zip Code

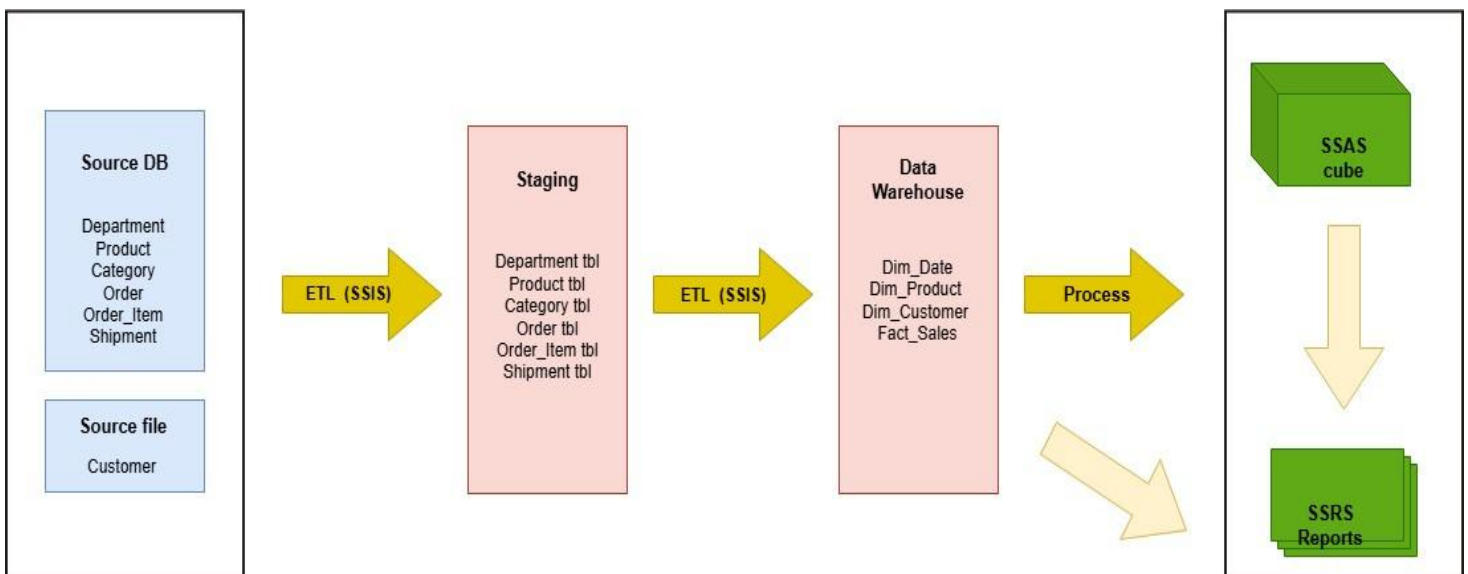
The file is formatted as a standard comma-separated CSV, allowing clean parsing during the ETL process using a Flat File Connection Manager in SSIS. This separation of customer data into a CSV demonstrates real-world integration of heterogeneous data sources.

Step 3: Solution Architecture

A high-level Data Warehouse (DW) and Business Intelligence (BI) solution architecture was designed to support the integration, transformation, and analysis of supply chain and logistics data for DataCo Global. The architecture reflects a realistic enterprise BI pipeline with multiple layers, starting from operational data sources and ending with user-facing analytical tools.

3.1 Architectural Diagram

The following diagram illustrates the key components of the DW & BI solution implemented for the DataCo Smart Supply Chain System.



3.2 Component Summary

Layer	Component	Description
Source Systems	DataCo Supply Chain Source DB (SQL Server)	The primary OLTP system containing normalized tables related to Department, Category, Product, Order, Order_Item, and Shipment.
	Customer Source File (CSV)	An external flat file that contains customer demographic and location data
Staging Area	DataCo_Staging	A temporary area where data from both the SQL database and CSV file is loaded, cleansed, and transformed before being moved to the data warehouse.
ETL Process	SSIS (SQL Server Integration Services)	Extracts data from heterogeneous sources, performs transformations, lookups, SCD Type 2 handling, and loads data into the data warehouse.
Data Warehouse	DataCo_DataWarehouse	A dimensional model implemented in SQL Server using a star schema. It consists of dimension tables (Dim_Date, Dim_Product, Dim_Customer) and one fact table (Fact_Sales).
BI Layer	SSAS Cube	Multidimensional cube built using SQL Server Analysis Services (SSAS), allowing OLAP queries and drill-down analytics.
	Power BI	Reporting services used to generate insightful reports and dashboards for supply chain performance, shipping efficiency, and late delivery risk analysis.

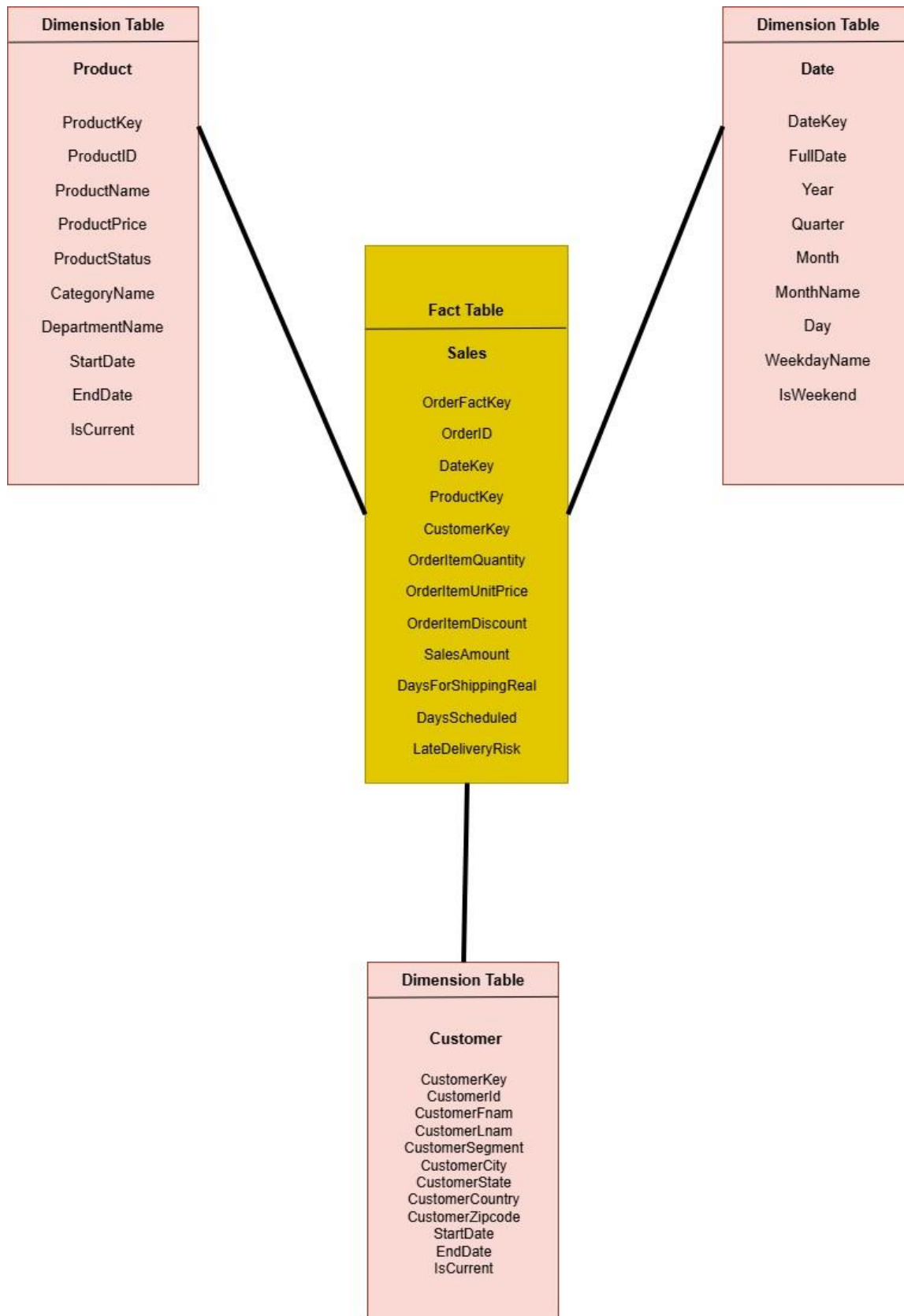
Step 4: Data Warehouse Design & Development

A star schema-based dimensional model was developed for the DataCo Smart Supply Chain system. This design provides an analytical foundation for exploring sales performance, product hierarchies, customer behavior, shipping efficiency, and late delivery risk over time. The schema includes one central fact table (Fact_Sales) and multiple related dimension tables, fulfilling the assignment requirements.

4.1 Dimensional Model Description

Table	Type	Description
Fact_Sales	Fact Table	Captures order-item level sales transactions including quantity, unit price, discount, sales amount, shipping performance metrics (real vs scheduled days), late delivery risk, and derived accumulating transaction processing time.
Dim_Product	Dimension	Contains product master data with hierarchies (Department → Category → Product). Implements SCD Type 2 logic to track changes in product price and status over time using StartDate and EndDate.
Dim_Customer	Dimension	Contains customer demographic and location information. Implements SCD Type 2 logic to track changes in customer segment, city, state, and country over time using StartDate and EndDate.
Dim_Date	Dimension	A conformed date dimension that enables time-based analysis by day, week, month, quarter, and year for orders and shipments.

4.2 Relational Diagram



4.3 Design Assumptions

- A **star schema** was chosen for its simplicity and performance in reporting scenarios.
- The **Fact_Sales** table includes one record per order-item combination.
- **Dim_Product** handles historical changes in product price and status using Slowly Changing Dimension (Type 2) logic.
- **Dim_Customer** handles historical changes in customer segment, city, state, and country using Slowly Changing Dimension (Type 2) logic.
- **Dim_Date** is a conformed dimension that supports time-based analysis for both order dates and shipment dates.
- **SalesAmount** and **txn_process_time_hours** are derived metrics created during the ETL process.
- Fact table links to dimensions via surrogate keys (e.g., **ProductKey**, **CustomerKey**, **DateKey**).
 - Product hierarchy (Department → Category → Product) is modeled through foreign key relationships across dimension tables.

Step 5: ETL Development

To populate the data warehouse with clean and integrated data, three ETL packages were developed using SQL Server Integration Services (SSIS). The ETL process is divided into three main stages:

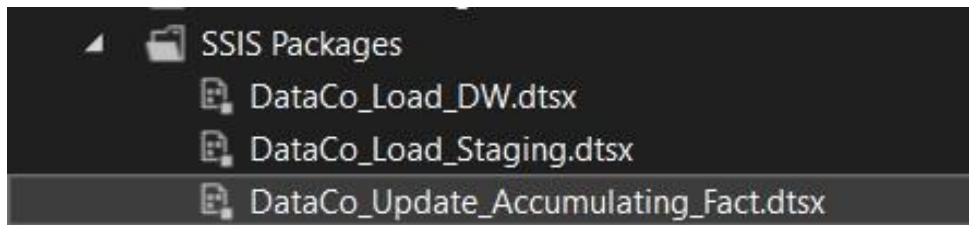
1. Data Extraction to Staging
2. Transformation and Loading to the Data Warehouse
3. Updating Accumulating Fact Table Attributes

Each package demonstrates key SSIS tasks such as connection setup, control and data flow design, transformations (Lookup, Derived Column, Data Conversion), SCD Type 2 handling, and load operations from multiple data sources (SQL Server database and CSV flat file).

5.1 SSIS Packages Overview

Three SSIS packages were created to organize the ETL workflow logically and modularly:

Package Name	Purpose
DataCo_Load_Staging.dtsx	Extracts data from the SQL Server source database (Department, Category, Product, Order, Order_Item, Shipment) and the external Customer_Source.csv file into staging tables.
DataCo_Load_DW.dtsx	Performs transformations and loads data into the dimension tables (Dim_Product and Dim_Customer using SCD Type 2) and the Fact_Sales fact table.
DataCo_Update_Accumulating_Fact.dtsx	Updates the accumulating fact attributes (accm_txn_complete_time and txn_process_time_hours) in the Fact_Sales table using late-arriving completion data from a separate CSV file.



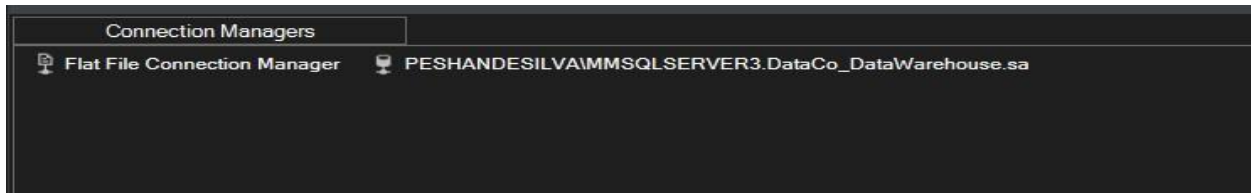
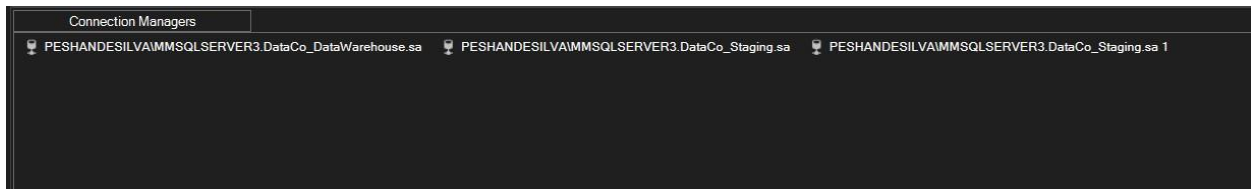
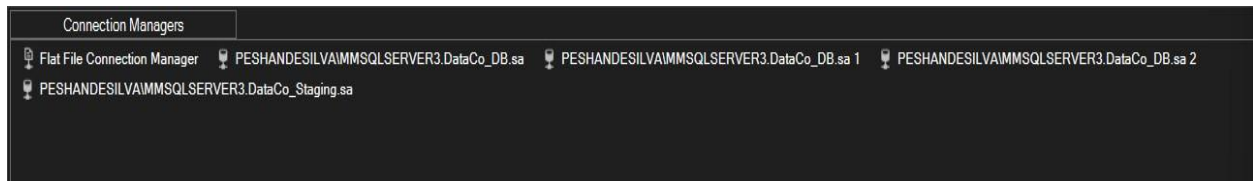
SSIS package modules as seen in Solution Explorer.

5.2 Connection Managers

All ETL packages utilize multiple connection types to handle data from both databases and CSV files. The following connections were configured:

- **SQL Server – OLE DB Connections:**
 - o DataCo_SupplyChain_SourceDB (source OLTP database)
 - o DataCo_Staging (staging database)
 - o DataCo_DataWarehouse (data warehouse)
- **Flat File Connections:**
 - o Customer_Source.csv (customer demographic and location data)
 - o Fact_Update.csv (for accumulating fact table updates)

These connection managers enable seamless integration between heterogeneous data sources.



5.3 Package 1: DataCo_Load_Staging.dtsx

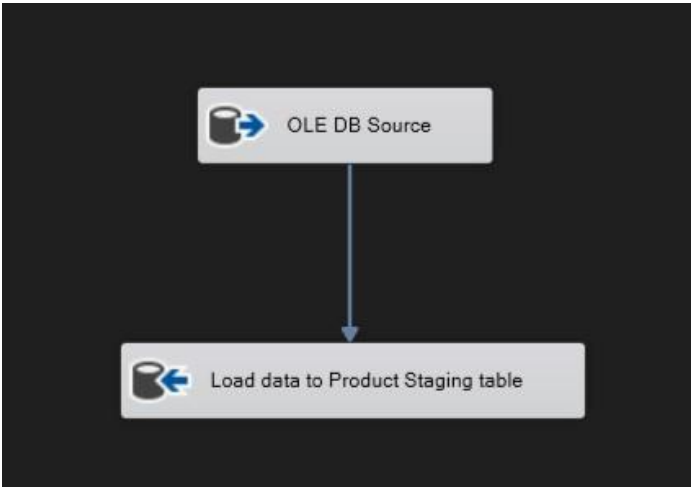
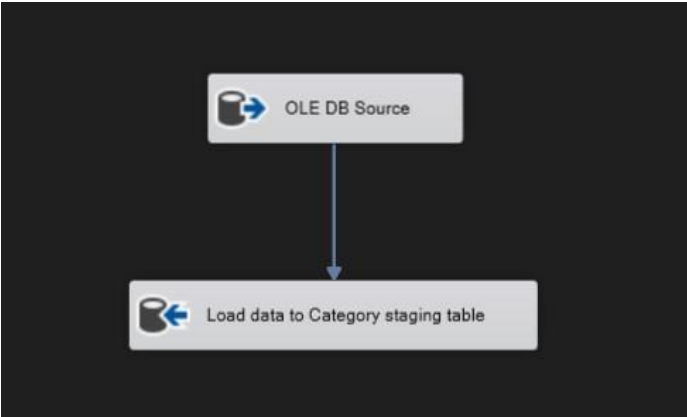
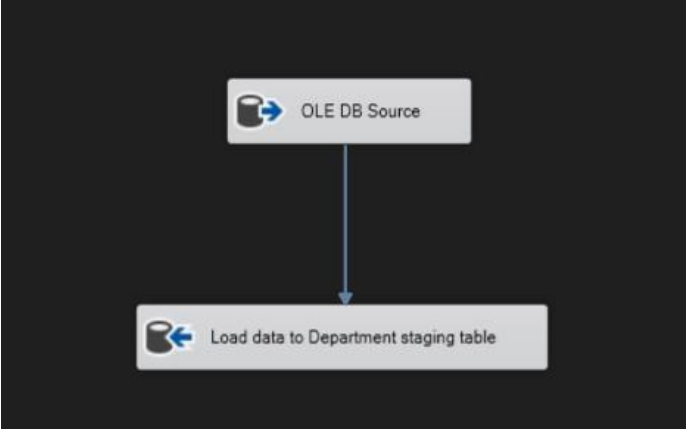
This package extracts raw data from source systems and loads it into staging tables. The control flow consists of the following sequential tasks:

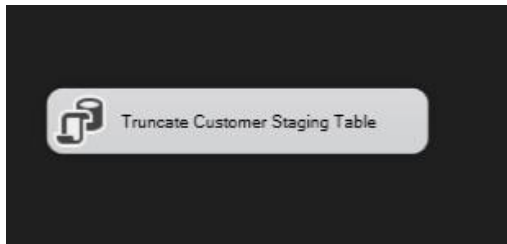
1. Extract Department to Staging
2. Extract Category to Staging
3. Extract Product to Staging
4. Extract Order to Staging

5. Extract Order_Item to Staging
6. Extract Shipment to Staging
7. Extract Customer to Staging (from the external CSV flat file)

Each task employs a simple OLE DB Source to OLE DB Destination (or Flat File Source to OLE DB Destination for the customer file where applicable).





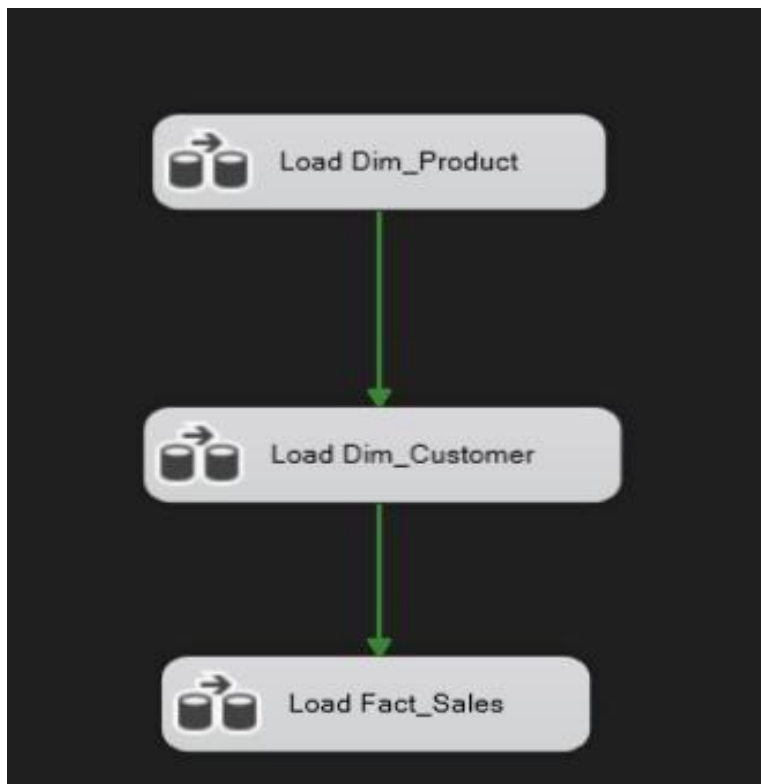


5.4 Package 2: DataCo_Load_DW.dtsx

This package handles the transformation and loading of clean data into the dimensional and fact tables of the data warehouse. The tasks are executed in the following order:

1. Transform and Load Product (into Dim_Product)
2. Transform and Load Customer (into Dim_Customer)
3. Transform and Load Sales (into Fact_Sales)

Within the Data Flow, surrogate keys are resolved using Lookup transformations (with IsCurrent = 1 filter for SCD Type 2), and derived attributes such as SalesAmount and txn_process_time_hours are computed using Derived Column transformations.



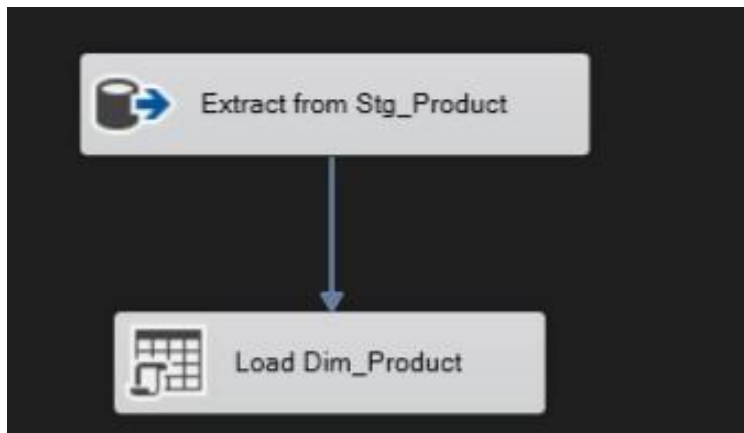
The image above shows the Control Flow of the second SSIS package (**DataCo_Load_DW.dtsx**). The three Execute Data Flow Tasks are executed sequentially to ensure proper loading order:

1. **Load Dim_Product** (SCD Type 2)

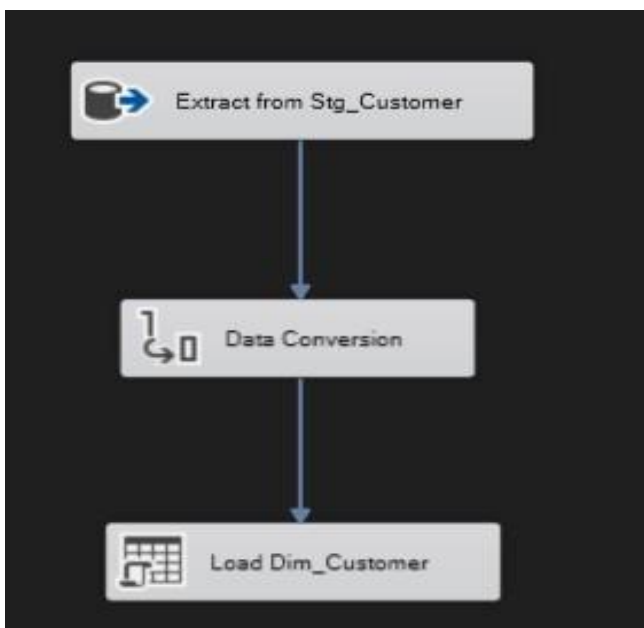
2. Load Dim_Customer (SCD Type 2)

3. Load Fact_Sales

This sequence guarantees that the dimension tables are fully populated with surrogate keys before the fact table is loaded. Inside each Data Flow task, Lookup transformations resolve surrogate keys from the dimensions, and Derived Column transformations are used to calculate measures such as SalesAmount.



Data is extracted from the staging table (Stg_Product) using an OLE DB Source and then loaded into the dimension table Dim_Product using an OLE DB Destination. This Data Flow implements SCD Type 2 logic (Slowly Changing Dimension) to track historical changes in product attributes such as price and status using StartDate and EndDate columns.



Data is extracted from the staging table (Stg_Customer) using an OLE DB Source. A Data Conversion transformation is applied to ensure data type compatibility between the source and destination. The data is then loaded into the Dim_Customer dimension table using an OLE DB Destination. This Data Flow implements SCD Type 2 logic (Slowly Changing Dimension) to track historical changes in customer attributes such as segment, city, state, and country over time using StartDate and EndDate columns.



Data is first extracted from the staging tables using a single OLE DB Source with a multi-table JOIN query (Stg_Order, Stg_Order_Item, and Stg_Shipment). Surrogate keys are then resolved using three Lookup transformations:

- Lookup ProductKey (from Dim_Product – SCD Type 2)
- Lookup CustomerKey (from Dim_Customer – SCD Type 2)
- Lookup DateKey (from Dim_Date)

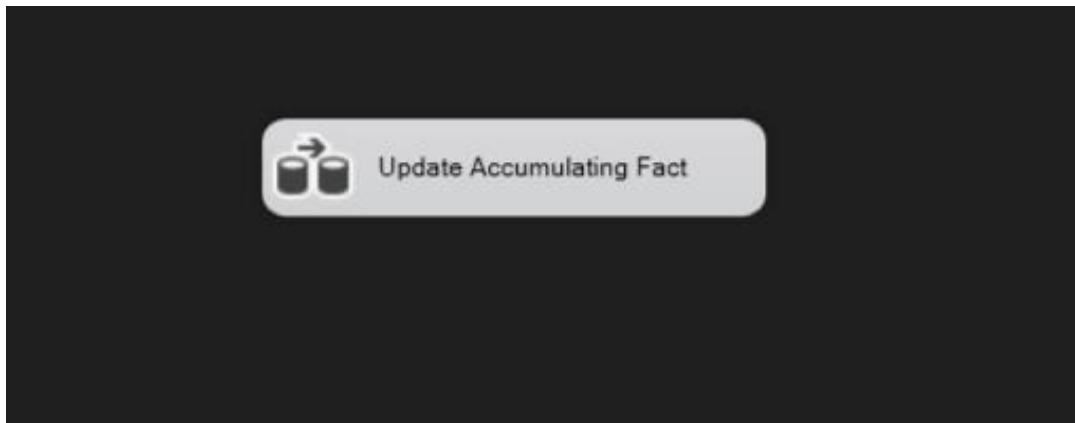
Derived Column transformations are used to calculate **SalesAmount** and to set **accm_txn_create_time = GETDATE()** for the accumulating fact table. Finally, the transformed data is loaded into the **Fact_Sales** table using an OLE DB Destination.

5.5 Package 3: DataCo_Update_Accumulating_Fact.dtsx

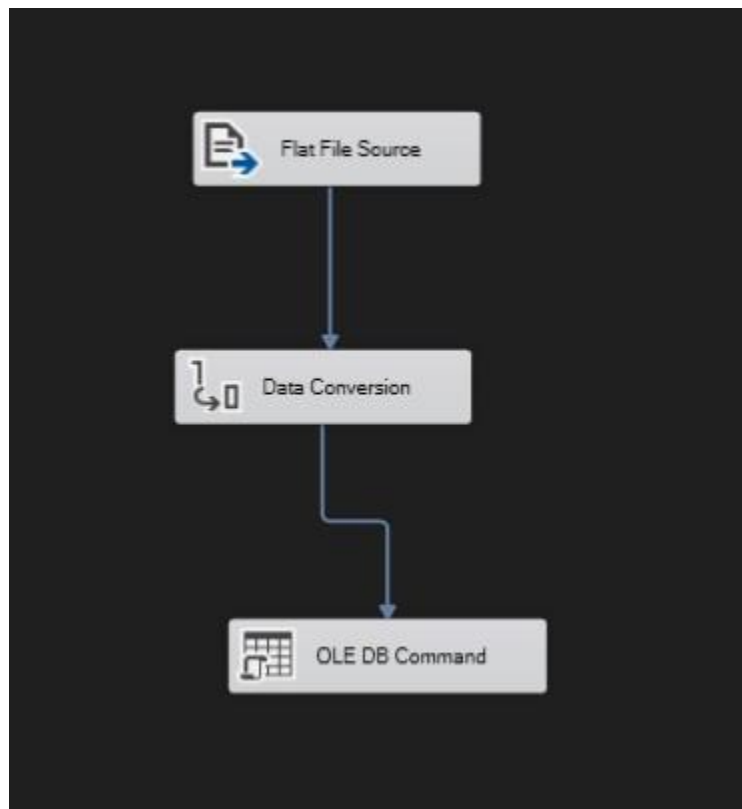
This package demonstrates the use of accumulating snapshot logic by updating an existing fact table using external completion data. The process includes:

1. **Stage Completed Transactions** – Load completion time data from a CSV file (Fact_Update.csv) into the ETL pipeline
2. **Update Fact_Sales** – Use an OLE DB Command transformation to update **accm_txn_complete_time** and calculate the derived field **txn_process_time_hours** using **DATEDIFF(HOUR, accm_txn_create_time, accm_txn_complete_time)**
3. **Clean up** – Ensures the update is applied only to matching OrderIDs while maintaining data integrity

This ensures historical accuracy while maintaining analytical relevance in the fact table.



5.5a Data Flow Task – DataCo_Update_Accumulating_Fact.dtsx



The screenshot above shows the complete Data Flow inside the third SSIS package (**DataCo_Update_Accumulating_Fact.dtsx**), which implements the accumulating snapshot fact table logic required in Task 6.

- **Flat File Source** reads the late-arriving transaction completion data from the external CSV file (Fact_Update.csv).
- **Data Conversion** transformation ensures proper data type compatibility (especially for the accm_txn_complete_time column).
- **OLE DB Command** performs a parameterized UPDATE on the Fact_Sales table. It updates accm_txn_complete_time and calculates the derived field txn_process_time_hours using the DATEDIFF(HOUR, ...) function.

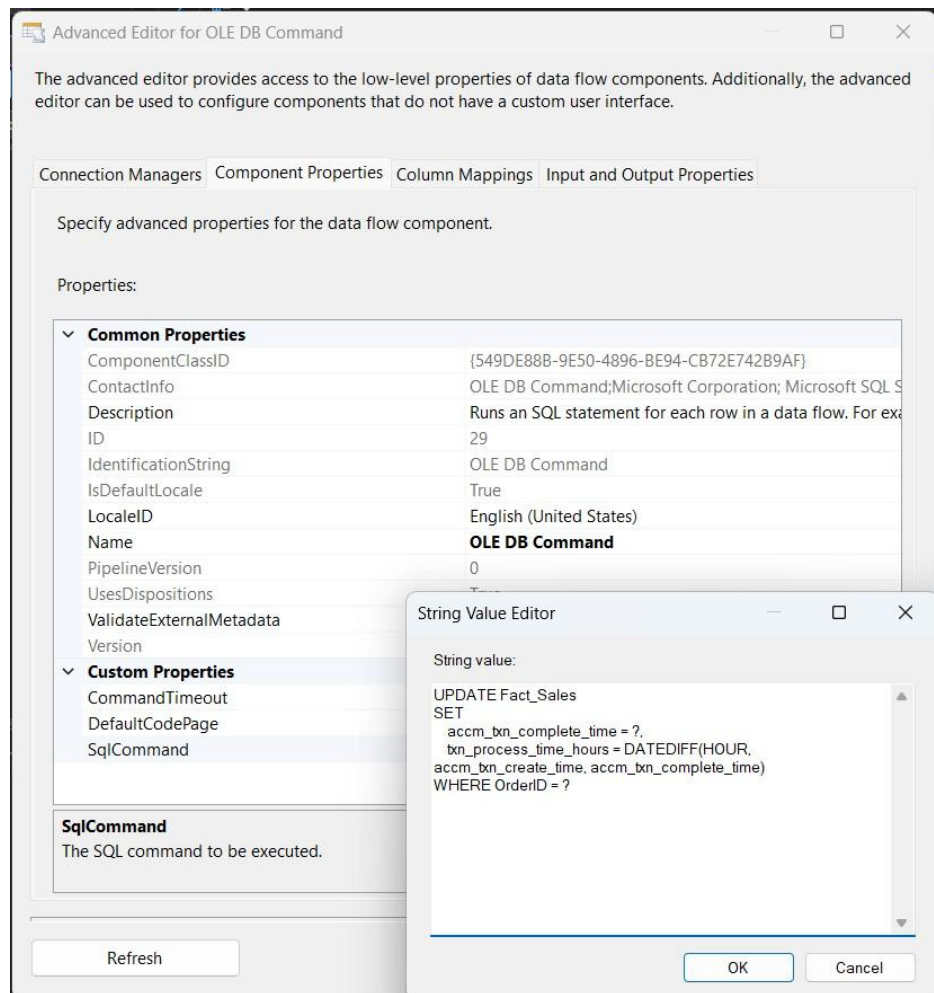
This design successfully handles late-arriving facts and keeps the fact table up-to-date while preserving historical accuracy.

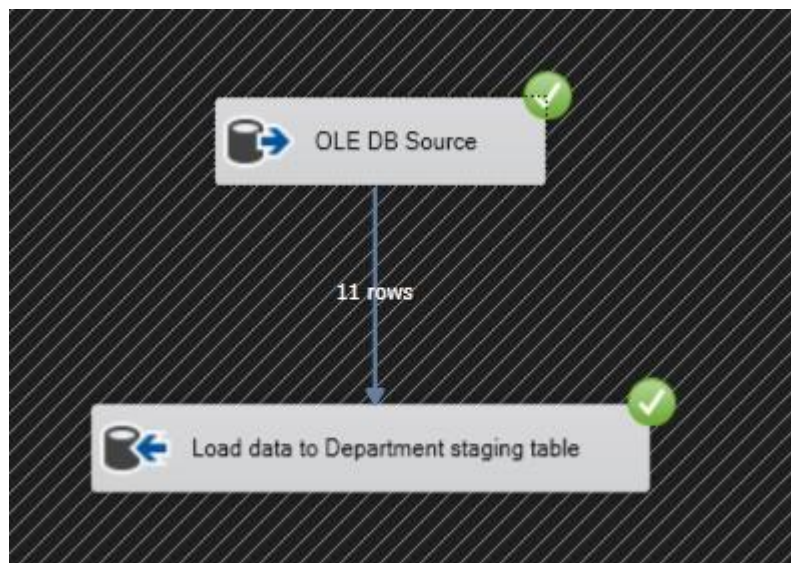
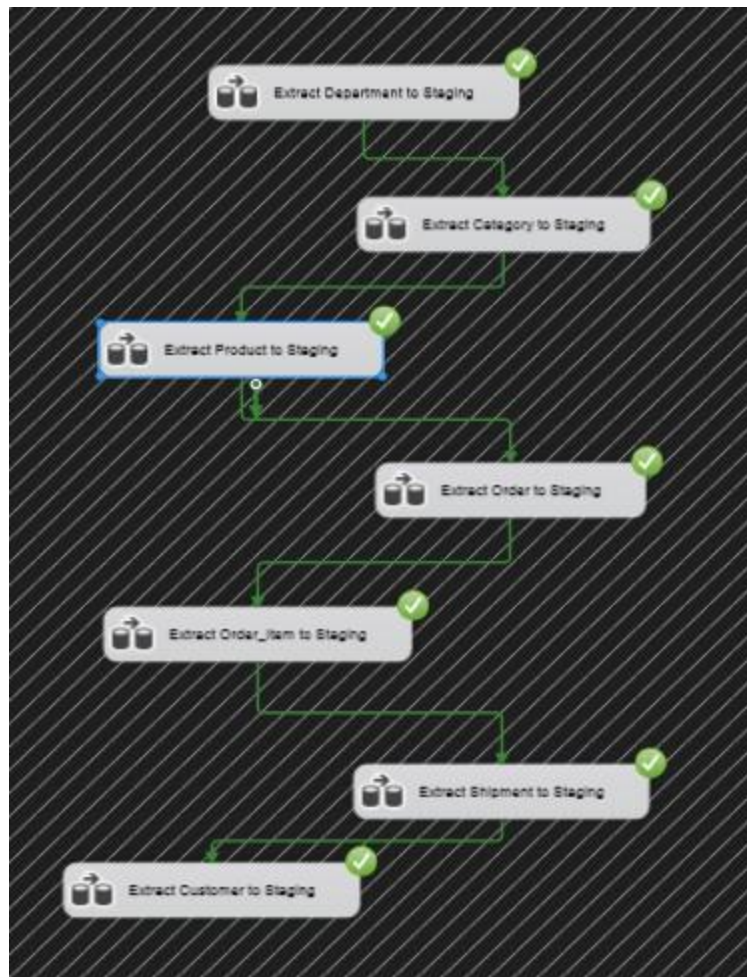
5.5b Fact Table Update – Completion Time and Process Duration

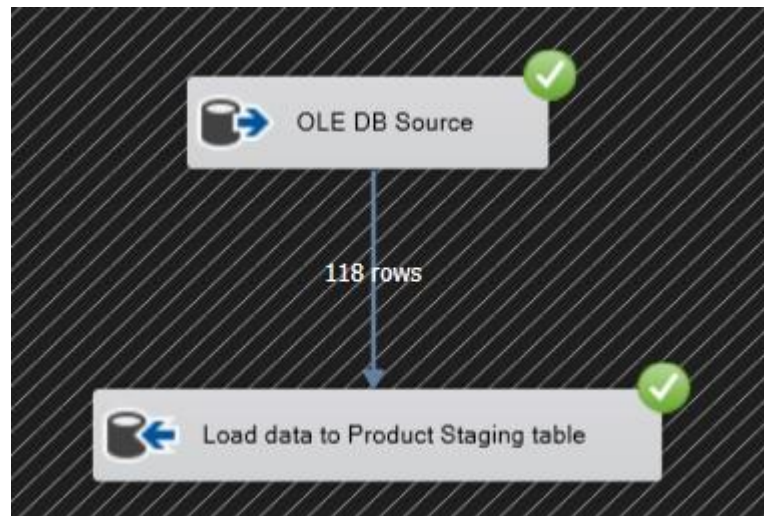
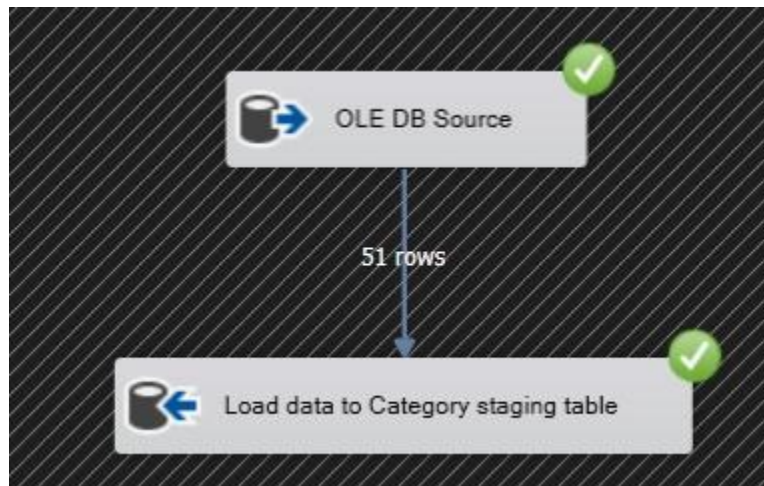
After reading the completed transaction data from the external CSV file (Fact_Update.csv), an **OLE DB Command** transformation in the **DataCo_Update_Accumulating_Fact.dtsx** package is used to update the accumulating snapshot fact table **Fact_Sales**. This update adds two important fields to the existing records:

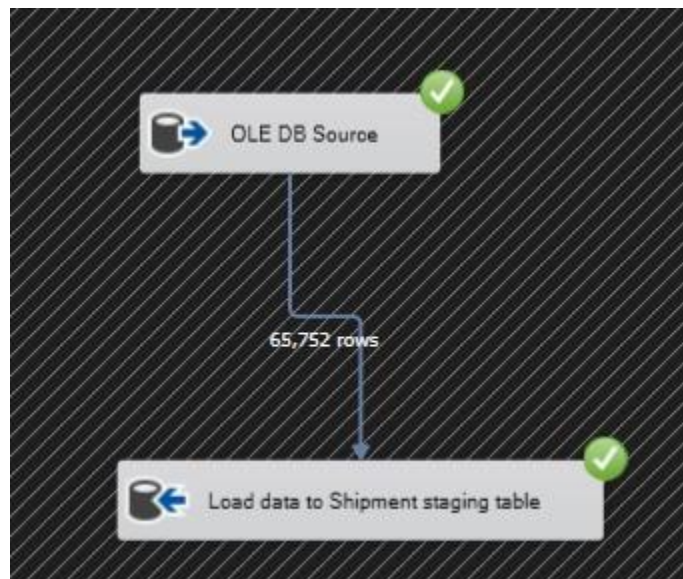
- **accm_txn_complete_time**: The exact timestamp when the order/transaction was marked as completed, sourced from the update CSV file.
- **txn_process_time_hours**: A derived metric that calculates the total processing time in hours between the initial creation of the transaction (accm_txn_create_time) and its completion. This is calculated using the SQL Server DATEDIFF(HOUR, accm_txn_create_time, accm_txn_complete_time) function.

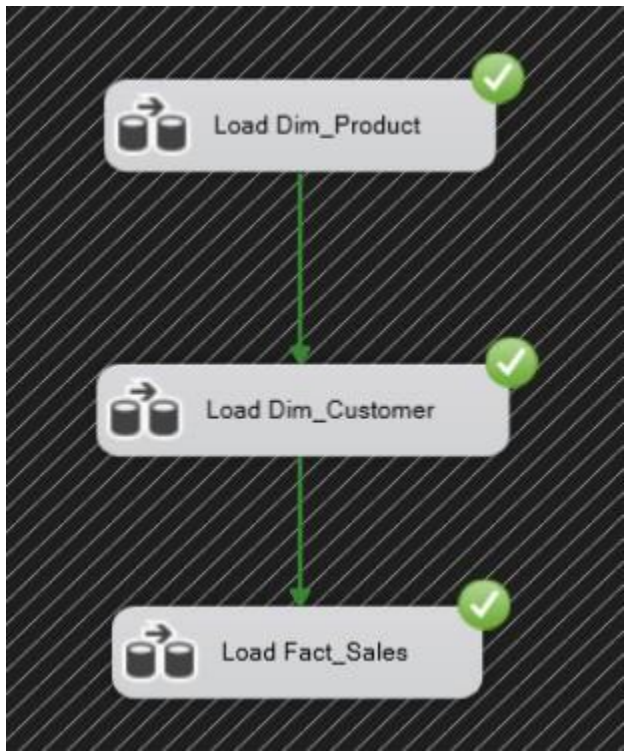
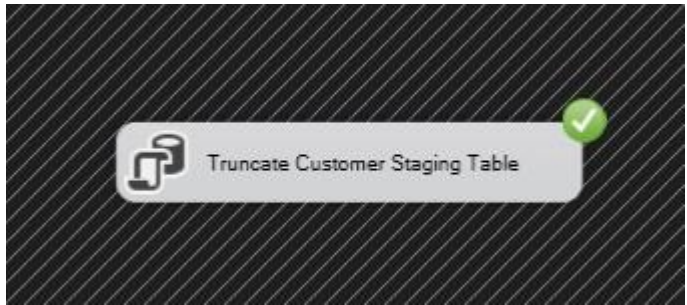
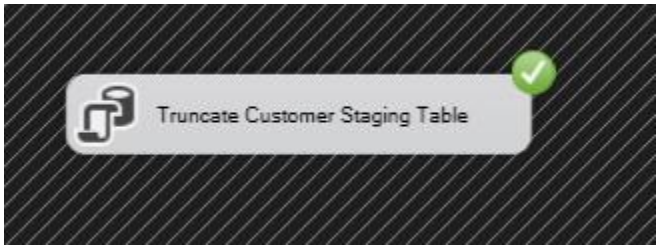
This update enriches the fact table with time-based performance indicators, enabling better analysis of order processing duration, supply chain efficiency, and late completion trends.

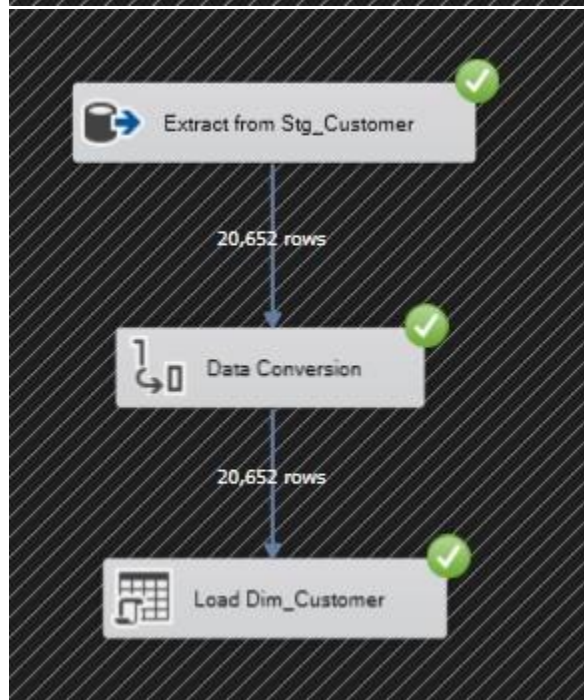


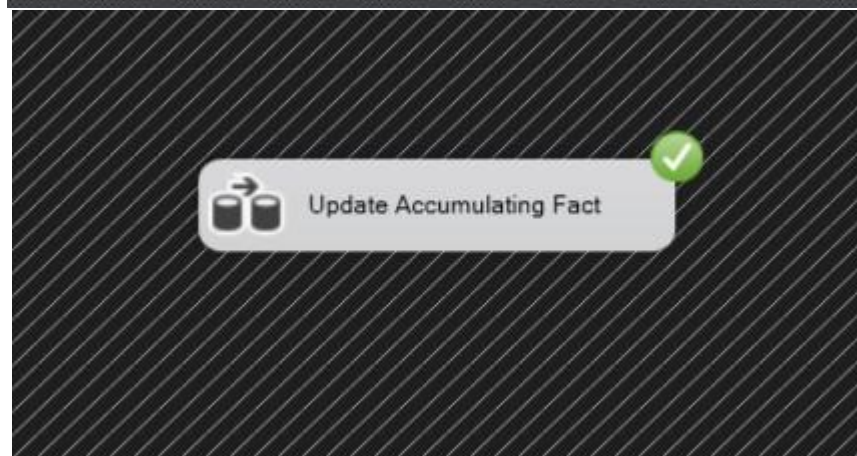
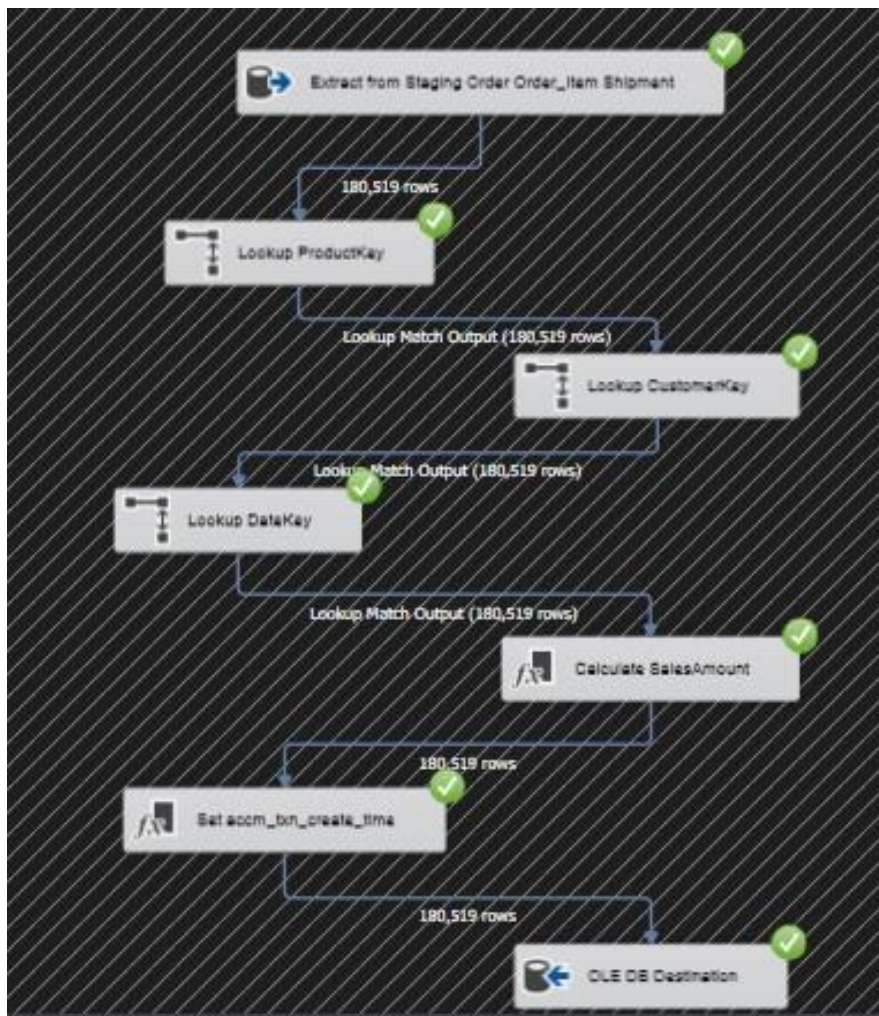


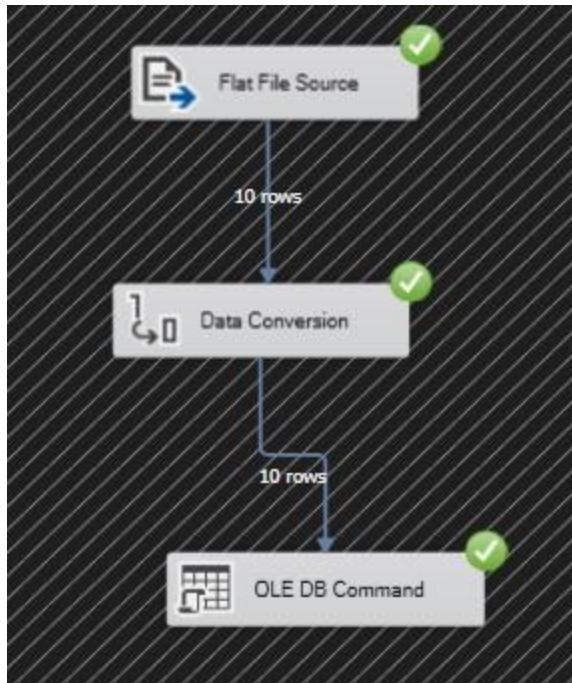












6.0 Conclusion

This assignment successfully demonstrated the design and implementation of a complete data warehouse solution for the **DataCo Smart Supply Chain** system. The project involved integrating data from a normalized OLTP database and an external CSV flat file, simulating a realistic enterprise supply chain environment.

The source data was extracted, staged, and transformed using SQL Server Integration Services (SSIS), with clearly defined control and data flow pipelines. The dimensional model was designed using a star schema consisting of one central fact table (**Fact_Sales**) and multiple related dimension tables (**Dim_Product**, **Dim_Customer**, and **Dim_Date**), allowing for efficient and scalable analytical querying.

Key data warehousing techniques such as Slowly Changing Dimensions (Type 2 for both Product and Customer), derived column transformations (e.g., SalesAmount), and accumulating snapshot logic (accm_txn_create_time, accm_txn_complete_time, and txn_process_time_hours) were successfully implemented. The modular ETL design, using three separate SSIS packages, improved maintainability and allowed for logical separation of data loading stages.

Although OLAP cube development and reporting are beyond the scope of this assignment, the data warehouse is fully compatible with SSAS and Power BI for future reporting and analysis.

Overall, this project provided hands-on experience in data integration, dimensional modeling, ETL development, Slowly Changing Dimensions, and accumulating fact table updates within a business intelligence context.

